



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no

Name

CNIC

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 04/06/2025 02:00 AM

Test Center No

Examiner Name

Signatures

Candidate Signature

Examiner Signature & Stamp

Principal Signature

Instructions:

Please read instructions carefully before attempting paper:

- 1) All questions are compulsory.
- 2) The question paper consists of 21 questions divided into two sections A & B. Section A comprise of 20 MCQs of 1 marks each. Section B comprises of 1 practical task having 80 marks.
- 3) Mobile Phone not allowed.
- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- When developing an application that requires understanding and processing natural language input, which Azure service provides a platform for building and managing language understanding models, enabling intent recognition and entity extraction?

- (a) Azure Language Understanding
- (b) Azure Content Moderator
- (c) Azure Text Analytics
- (d) Azure Translator

2- In deep learning, which type of neural network is commonly used for image classification tasks, leveraging 2D convolutional layers to automatically learn hierarchical features from images?

- (a) 2D Convolutional Neural Network (CNN)
- (b) Autoencoder
- (c) Multi-Layer Perceptron (MLP)
- (d) Recurrent Neural Network (RNN)

3- In deep learning, what type of neural network architecture consists of multiple layers of interconnected neurons, including input, hidden, and output layers, where information flows only in one direction from input to output?

- (a) Recurrent Neural Network (RNN)
- (b) Long Short-Term Memory (LSTM)
- (c) MLP Feed Forward Neural Network
- (d) Convolutional Neural Network (CNN)

4- What is dynamic polymorphism in object-oriented programming?

- (a) A technique to combine multiple classes into one
- (b) The ability of a subclass to override methods of its base class
- (c) A way to prevent inheritance between classes
- (d) The process of defining new data types

5- In object-oriented programming, what is multilevel inheritance?

- (a) Inheriting from multiple parent classes
- (b) Inheriting from a child class
- (c) Inheriting from a grandparent class
- (d) Inheriting from a single parent class

6- Which service can you use to analyze images and extract text from them in Microsoft Azure?

- (a) Anomaly Detector
- (b) Personalizer
- (c) Metrics Advisor
- (d) Content Moderator

7- In logistic regression, what type of output does the model produce?

- (a) Probability scores
- (b) Continuous numerical values
- (c) Categorical labels
- (d) Clusters of data points

8- Which command is used to display the contents of a directory in the terminal?

- (a) ls
- (b) display
- (c) dir
- (d) show

9- What are universal array functions in Numpy?

- (a) Functions that perform matrix multiplication
- (b) Functions that operate element-wise on arrays
- (c) Functions that only work on 1D arrays
- (d) Functions that require external libraries

10- In machine learning, which ensemble method combines multiple decision trees by training them on different subsets of the training data and averaging their predictions to improve model accuracy and reduce overfitting?

- (a) Random Forest
- (b) K-Means Clustering
- (c) Gradient Boosting
- (d) AdaBoost

11- When planning for an AI solution that aligns with Responsible AI principles, which practice involves ensuring transparency and explainability of AI models and decisions, enabling users to understand the reasoning behind predictions?

- (a) Model Accuracy
- (b) Explainability and Transparency
- (c) Performance Metrics
- (d) Data Collection

12- In object-oriented programming, what are class variables?

- (a) Variables that are shared among all instances of a class
- (b) Variables that belong to the class itself
- (c) Variables that store data unique to each instance of a class
- (d) Variables used for method definition

13- In univariate linear regression without vectorization, how is the gradient of the cost function calculated for the slope parameter?

- (a) Using matrix multiplication
- (b) Using a direct formula
- (c) Using a loop to iterate over each data point
- (d) By taking the derivative of the cost function

14- In neural networks, what type of functions are applied to the output of neurons to introduce nonlinearity and enable the network to learn complex relationships in data?

- (a) Gradient functions
- (b) Activation functions
- (c) Loss functions
- (d) Regularization functions

15- In statistics, which term describes a variable that can take on distinct, separate values?

- (a) Quantitative
- (b) Qualitative
- (c) Discrete
- (d) Continuous

16- How can you save a Numpy array to a file?

- (a) numpy.save()
- (b) numpy.store()
- (c) numpy.write()
- (d) numpy.export()

17- What is the primary application of Convolutional Neural Networks (CNNs)?

- (a) Tabular data analysis
- (b) Text classification
- (c) Image and video analysis
- (d) Speech recognition

18- When creating an Azure AI resource, which service in the Azure portal allows you to provision and manage AI capabilities, such as cognitive services and machine learning models, for your applications?

- (a) Azure App Service
- (b) Azure Storage
- (c) Azure Virtual Machine
- (d) Azure Cognitive Services

19- What field of study focuses on enabling computers to understand, interpret, and generate human language?

- (a) Computer Vision
- (b) Data Science
- (c) Natural Language Processing
- (d) Machine Learning

20- Which measure provides information about the relative position of a data point within a dataset?

- (a) Mean
- (b) Standard Deviation
- (c) Percentile
- (d) Range

__The End__



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Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- In Python, which library provides an easy-to-use interface for building and training neural networks, and offers a high-level API for defining layers and model architectures?

- (a) Keras
- (b) TensorFlow
- (c) Scikit-learn
- (d) PyTorch

2- Which type of loop is suitable when the number of iterations is known beforehand?

- (a) While loop
- (b) Do-while loop
- (c) For loop
- (d) Infinite loop

3- What is the purpose of using keyword arguments in function calls?

- (a) To increase the performance of a function
- (b) To assign a name to an argument
- (c) To change the order of arguments
- (d) To make a function private

4- In data analysis, what process involves converting continuous numerical data into discrete intervals or bins?

- (a) Binning
- (b) Discretization
- (c) Categorization
- (d) Segmentation

5- In univariate linear regression without vectorization, how is the gradient of the cost function calculated for the slope parameter?

- (a) Using matrix multiplication
- (b) Using a direct formula
- (c) Using a loop to iterate over each data point
- (d) By taking the derivative of the cost function

6- When building a solution that provides personalized recommendations and content for users, which Azure Cognitive Service is designed to understand user preferences and deliver customized experiences based on real-time interaction data?

- (a) Azure Text Analytics
- (b) Azure Personalizer
- (c) Azure Speech Service
- (d) Azure Translator

7- If you're building a speech solution that involves converting spoken language into text, performing text-to-speech synthesis, and enabling speaker recognition, which Azure Cognitive Services offering is best suited for these tasks?

- (a) Azure Text Analytics
- (b) Azure Language Understanding
- (c) Azure Speech Service
- (d) Azure Translator

8- What is the process of breaking a text into individual words or tokens?

- (a) Lemmatization
- (b) Sentence tokenization
- (c) Word tokenization
- (d) Text normalization

9- Which command is used to display the contents of a file in a Unix-like operating system?

- (a) read
- (b) view
- (c) cat
- (d) open

10- What technique is used in Pandas to handle missing data in a DataFrame?

- (a) Data encryption
- (b) Data imputation
- (c) Data dropping
- (d) Data shuffling

11- Which measure provides information about the relative position of a data point within a dataset?

- (a) Mean
- (b) Standard Deviation
- (c) Percentile
- (d) Range

12- In Pandas, which data structure is used to store one-dimensional labeled data, similar to a column in a spreadsheet?

- (a) List
- (b) DataFrame
- (c) Array
- (d) Series

13- What is the advantage of using functions in programming?

- (a) They allow you to define new data types
- (b) They help you optimize memory usage
- (c) They allow you to create GUI interfaces
- (d) They allow you to reuse code and improve code organization

14- In object-oriented programming, what are instance variables?

- (a) Variables that are shared among all instances of a class
- (b) Variables that store data unique to each instance of a class
- (c) Variables that belong to the class itself
- (d) Variables used for method definition

15- In languages without explicit spaces between words, what is the process of dividing a sequence of characters into individual words called?

- (a) Sentence tokenization
- (b) Part-of-speech tagging
- (c) Text normalization
- (d) Word segmentation

16- What is the main goal of machine learning?

- (a) To program computers to think like humans
- (b) To enable computers to learn from data and improve performance over time
- (c) To automate tasks without using data
- (d) To optimize computer hardware

17- In natural language processing, what is the process of reducing a word to its base or root form called, by removing suffixes or prefixes?

- (a) Segmentation
- (b) Stemming
- (c) Lemmatization
- (d) Tokenization

18- Which measures assess the spread or dispersion of data points around the mean?

- (a) Mean and Standard Deviation
- (b) Mean and Median
- (c) Variance and Standard Deviation
- (d) Range and Mode

19- What is a lexical database in NLP that provides a semantic network of words and their relationships, including synonyms and hypernyms?

- (a) NER tagging
- (b) Stemming
- (c) POS tagging
- (d) WordNet

20- What field of study focuses on enabling computers to understand, interpret, and generate human language?

- (a) Computer Vision
- (b) Data Science
- (c) Natural Language Processing
- (d) Machine Learning

__The End__